MATHEMATICS STANDARD
CBSE Examination Paper-2024

With Topper's Answer

Time allowed: 3 hours *Maximum Marks: 80*

General Instructions:

Read the following instructions very carefully and strictly follow them:

*(i) This question paper contains **38** questions. **All** questions are **compulsory.***

*(ii) This question paper is divided into **Five** Sections - **A, B, C, D** and **E**.*

*(iii) In **Section A** questions number **1** to **18** are Multiple Choice Questions (MCQs) and question number **19** and **20** are Assertion-Reason based questions of **1** mark each.*

*(iv) In **Section B** questions number **21** and **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.*

*(v) In **Section C** questions number **26** to **31** are Short Answer (SA) type questions, carrying **3** marks each.*

*(vi) In **Section D** questions number **32** to **35** are Long Answer (LA) type questions, carrying **5** marks each.*

*(vii) In **Section E** questions number **36** to **38** are Case Study based questions carrying **4** marks each. Internal choice is provided in **2** marks questions in each case-study.*

*(viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section-**B**, 2 questions in Section-**C**, 2 questions in Section-**D** and 3 question in Section-**E**.*

(ix) Draw neat diagrams wherever required, Take $\pi = \frac{22}{7}$ wherever required, if not stated.

*(x) Use of calclator is **NOT** allowed.*

SECTION - A **20×1=20**

This section consists of 20 questions of 1 mark each.

1. The value of k for which the system of equations $3x - y + 8 = 0$ and $6x - ky + 16 = 0$ has infinitely many solutions, is **1**

(A) –2 **(B)** 2 **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{2}$

Ans. (B) 2.

2. Point P divides the line segment joining the points A(4, –5) and B(1, 2) in the ratio 5:2. Co-ordinates of point P are **1**

(A) $\left(\frac{5}{2}, \frac{-3}{2}\right)$ **(B)** $\left(\frac{11}{7}, 0\right)$ **(C)** $\left(\frac{13}{7}, 0\right)$ **(D)** $\left(0, \frac{13}{7}\right)$

Ans. (C) $\left(\frac{13}{7}, 0\right)$

3. The common difference of an A.P. in which $a_{15} - a_{11} = 48$, is **1**

(A) 12 **(B)** 16 **(C)** –12 **(D)** –16

Ans. (A) 12

4. The quadratic equation $x^2 + x + 1 = 0$ has _____ roots. **1**

(A) real and equal **(B)** irrational **(C)** real and distinct **(D)** not-real

Ans. (D) not-real

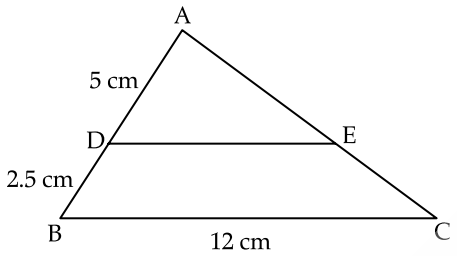
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5. If the HCF (2520, 6600) = 40 and LCM (2520, 6600) = 252 × k, then the value of k is **1**

(A) 1650 **(B)** 1600 **(C)** 165 **(D)** 1625

Ans. (A) 1650

6. In the given figure ΔABC is shown. DE is parallel to BC. If AD = 5 cm, DB = 2.5 cm and BC = 12 cm, then DE is equal to **1**



(A) 10 cm **(B)** 6 cm **(C)** 8 cm **(D)** 7.5 cm

Ans. (C) 8cm

7. If sin θ = cos θ, (0° < θ < 90°), then value of (sec θ . sin θ) is : **1**

(A) $\frac{1}{\sqrt{2}}$ **(B)** $\sqrt{2}$ **(C)** 1 **(D)** 0

Ans. (C) 1

8. Two dice are rolled together. The probability of getting the sum of the two numbers to be more than 10, is **1**

(A) $\frac{1}{9}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{7}{12}$ **(D)** $\frac{1}{12}$

Ans. D) $\frac{1}{12}$.

9. If α and β are zeroes of the polynomial $5x^2 + 3x - 7$, the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is **1**

(A) $-\frac{3}{7}$ **(B)** $\frac{3}{5}$ **(C)** $\frac{3}{7}$ **(D)** $-\frac{5}{7}$

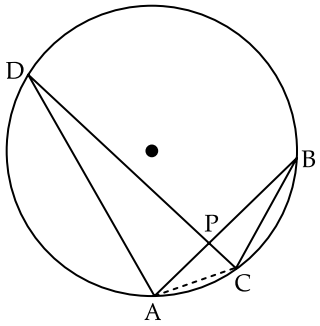
Ans. (C) $\frac{3}{7}$

10. The perimeters of two similar triangles ABC and PQR are 56 cm and 48 cm respectively. PQ/AB is equal to **1**

(A) $\frac{7}{8}$ **(B)** $\frac{6}{7}$ **(C)** $\frac{7}{6}$ **(D)** $\frac{8}{7}$

Ans. (B) $\frac{6}{7}$

11. AB and CD are two chords of a circle intersecting at P. Choose the correct statement from the following: **1**



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(A) $\Delta ADP \sim \Delta CBA$ **(B)** $\Delta ADP \sim \Delta BPC$ **(C)** $\Delta ADP \sim \Delta BCP$ **(D)** $\Delta ADP \sim \Delta CBP$

Ans. (D) $\Delta ADP \sim \Delta CBP$

12. If value of each observation in a data is increased by 2, then median of the new data **1**

(A) increases by 2 **(B)** increases by 2n **(C)** remains same **(D)** decreases by 2

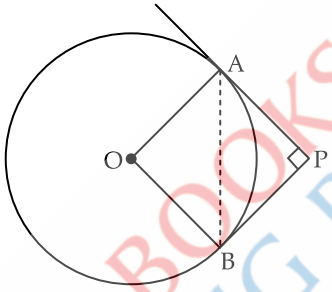
Ans. (A) increases by 2.

13. A box contains cards numbered 6 to 55. A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square, is **1**

(A) $\frac{7}{50}$ **(B)** $\frac{7}{55}$ **(C)** $\frac{1}{10}$ **(D)** $\frac{5}{49}$

Ans. (C) $\frac{1}{10}$

14. In the given figure, tangents PA and PB to the circle centred at O, from point P are perpendicular to each other. If PA = 5 cm, then length of AB is equal to **1**



(A) 5 cm **(B)** $5\sqrt{2}$ cm **(C)** $2\sqrt{5}$ cm **(D)** 10 cm

Ans. (B) $5\sqrt{2}$ cm

15. XOYZ is a rectangle with vertices X(–3, 0), O(0, 0), Y(0, 4) and Z(x, y). The length of its each diagonal is **1**

(A) 5 units **(B)** $\sqrt{5}$ units **(C)** $x^2 + y^2$ units **(D)** 4 units

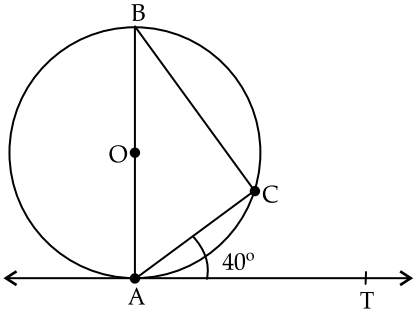
Ans. (A) 5 units

16. Which term of the A.P. –29, –26, –23, ..., 61 is 16 ? **1**

(A) 11th **(B)** 16th **(C)** 10th **(D)** 31st

Ans. (B) 16th

17. In the given figure, AT is tangent to a circle centred at O. If $\angle CAT = 40°$, then $\angle CBA$ is equal to **1**



(A) 70° **(B)** 50° **(C)** 65° **(D)** 40°

Ans. (D) 40°

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18. After an examination, a teacher wants to know the marks obtained by maximum number of the students in her class. She requires to calculate _____ of marks. **1**

(A) median **(B)** mode **(C)** mean **(D)** range

Ans. (B) mode

Directions: In Question **19** and **20**, **Assertion (A)** and **Reason (R)** are given. Select the correct option from the following :

(A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true. Reason (R) does not give correct explanation of (A).

(C) Assertion (A) is true but Reason (R) is not true.

(D) Assertion (A) is not true but Reason (R) is true.

19. **Assertion (A):** If $\sin A = \frac{1}{3}$ $(0° < A < 90°)$, then the value of $\cos A$ is $\frac{2\sqrt{2}}{3}$

Reason (R): For every angle θ, $\sin^2\theta + \cos^2\theta = 1$. **1**

Ans. (A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

20. **Assertion (A):** Two cubes each of edge length 10 cm are joined together. The total surface area of newly formed cuboid is 1200 cm^2.

Reason (R): Area of each surface of a cube of side 10 cm is 100 cm^2. **1**

Ans. (D) Assertion (A) is not true but Reason (R) is true.

SECTION - B

In this section, there are 5 questions of 2 marks each.

21. Can the number $(15)^n$, n being a natural number, end with the digit 0 ? Give reasons. **2**

Ans. If the number $(15)^n$, n being a natural number, were to end with the digit 0, then it would be divisible by 2 and 5.
This means it must have 2 among its prime factors.
However $(15)^n$ can only be factorised as $(3)^n \times (5)^n = (3\times5)^n$.
It does not have 2 among its factors.
The uniqueness of the 'Fundamental Theorem of Arithmetic' guarantees that the composition of 15 has only the prime nos 3 and 5.
Hence $(15)^n$ cannot end with the digit zero ever.

22. Find the type of triangle ABC formed whose vertices are A(1, 0), B(–5, 0) and C(–2, 5). **2**

Ans. Given : A(1, 0) ; B(–5, 0) ; C(–2, 5)
To find : Type of triangle formed
Solution : Using the distance formula ie, $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
we find the distance between each side.

$$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$= \sqrt{(1+5)^2+(0-0)^2}$$
$$= \sqrt{36}$$
$$= 6 \text{ units}$$

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$BC = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
$= \sqrt{(-5+2)^2+(0-5)^2}$
$= \sqrt{(-3)^2+(-5)^2} \Rightarrow \sqrt{9+25}$
$= \sqrt{34}$ units

$CA = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
$= \sqrt{(1+2)^2+(0-5)^2}$
$= \sqrt{(3)^2+(5)^2} \Rightarrow \sqrt{9+25}$
$= \sqrt{34}$ units

Since $\triangle ABC$ has 2 sides (CA and BC) of equal length (each $\sqrt{34}$ units)

we can conclude that $\triangle ABC$ is isosceles.

$\therefore \triangle ABC$ is an isosceles triangle.

23. (a) Evaluate : $2 \sin^2 30° \sec 60° + \tan^2 60°$. **2**

OR

(b) If $2 \sin (A + B) = \sqrt{3}$ and $\cos (A - B) = 1$, then find the measures of angles A and B. $0 \le A, B, (A + B) \le 90°$. **2**

Ans. (b) Given : $2\sin(A+B) = \sqrt{3}$ and $\cos(A-B) = 1$

To find : values of A and B

Solution : Consider $2\sin(A+B) = \sqrt{3}$

$\Rightarrow \sin(A+B) = \sqrt{3}/2$

$\Rightarrow \sin(A+B) = \sin(60°)$

$A + B = 60° \longrightarrow (1)$

Also it is given, $\cos(A-B) = 1$

$\Rightarrow \cos(A-B) = \cos(0°)$

$\Rightarrow A - B = 0 \longrightarrow (2)$

now adding (1) and (2), $A + B + A - B = 60° + 0°$

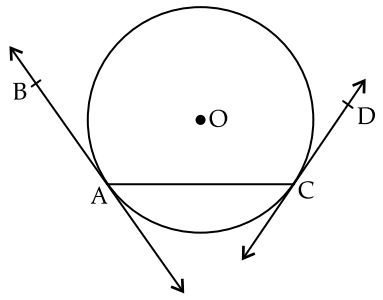
$2A = 60°$

$\therefore A = 30°$

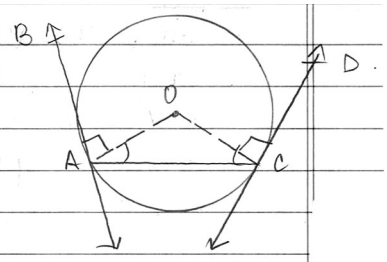
substituting value of A in (1), we get $B = 30°$

Thus values of A and B are 30° and 30° respectively.

24. In the given figure, AB and CD are tangents to a circle centred at O. Is $\angle BAC = \angle DCA$? Justify your answer. **2**



Ans.



Given: AB and CD are tangents to circle with centre O and chord AC.

To find: if $\angle BAC = \angle DCA$

Construction: Join OA and OC

Solution: Consider $\triangle OAC$ where

$OA = OC$ (radii of same circles)

so $\angle OAC = \angle OCA \rightarrow (1)$

(since angles opposite to equal sides in a triangle are equal)

We also know that a tangent to a circle is perpendicular to radius at the point of contact. Since AB and CD are both tangents,

$\therefore \angle OAB = \angle OCD \rightarrow (2)$ (as each is equal to 90°)

adding (1) and (2), $\angle OAC + \angle OAB = \angle OCA + \angle OCD$

$\angle BAC = \angle DCA$

hence proved that $\angle BAC = \angle DCA$.

25. (a) In what ratio is the line segment joining the points (3, –5) and (–1, 6) divided by the line $y = x$? **2**

Ans. (a) Let the points be A(3,–5) and B(–1, 6) divided by the point C(x, y) such that $x = y$.

section formula gives us the coordinates for points divided =

$$(m:n) = \left[\frac{(m)(x_2)+(n)(x_1)}{m+n}, \frac{(m)(y_2)(n)(y_1)}{m+n}\right]$$

$$\Rightarrow (K:1) = \left[\frac{(K)(-1)+(1)(3)}{K+1}, \frac{(K)(+6)+(1)(-5)}{K+1}\right]$$

$$\Rightarrow (x,y) = \left[\frac{-K+3}{K+1}, \frac{+6K-5}{K+1}\right]$$

thus x coordinate $= \left(\frac{-K+3}{K+1}\right)$ and y coordinate $= \left(\frac{+6K-5}{K+1}\right)$

but we know that $x = y$.

$$\frac{-K+3}{K+1} = \frac{+6K-5}{K+1}$$

$$3+5 = 6K+K$$

$$8 = 7K$$

$$K = \frac{8}{7}.$$

hence the ratio is K:1 ie, 8:7

Required ratio is 8:7.

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OR

(b) A(3, 0), B(6, 4) and C(–1, 3) are vertices of a triangle ABC. Find length of its median BE. **2**

SECTION - C

This section consists of 6 questions of 3 marks each.

26. (a) If the sum of first m terms of an A.P. is same as sum of its first n terms ($m \neq n$), then show that the sum of its first (m + n) terms is zero. **3**

Ans. (a) Given : $S_m = S_n$

To prove : $S_{(m+n)} = 0$

Proof : We know that sum of first 'x' terms in AP is given by –

$$\boxed{S_x = \frac{x}{2}[2a + (x-1)d]}$$

so $S_m = \frac{m}{2}[2a + (m-1)d]$

and $S_n = \frac{n}{2}[2a + (n-1)d]$.

but $S_m = S_n$ (given)

$\Rightarrow \frac{m}{2}[2a + (m-1)d] = \frac{n}{2}[2a + (n-1)d]$

$\Rightarrow m[2a + md - d] = n[2a + nd - d]$.

$\Rightarrow 2am + m^2d - dm = 2an + n^2d - dn$

$\Rightarrow 2am - 2an + m^2d - n^2d - dm + dn = 0$

$\Rightarrow 2a(m-n) + d(m^2 - n^2) - d(m-n) = 0$

$\Rightarrow (m-n)[2a + d(m+n) - d] = 0$

$\Rightarrow (m-n)[2a + d(m+n-1)] = 0$.

$\Rightarrow [2a + d(m+n-1)] = 0$ — (1)

now sum of first (m+n) terms will be,

$S_{(m+n)} = \frac{(m+n)}{2}[2a + (m+n-1)d]$.

but using (1) we get,

$S_{(m+n)} = \frac{(m+n)}{2}[0]$

thus $S_{(m+n)} = 0$

hence proved that sum of first (m+n) terms of this AP will definitely be zero.

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OR

(b) In an A.P., the sum of three consecutive terms is 24 and the sum of their squares is 194. Find the numbers. **3**

27. Prove that $\sqrt{5}$ is an irrational number. **3**

Ans

Let us assume to the contrary that $\sqrt{5}$ is rational. Then it can be expressed in the form (a/b) where 'a' and 'b' are integers and co-primes. Also, $b \neq 0$.

So, $\sqrt{5} = \frac{a}{b}$

$\Rightarrow$ $b\sqrt{5} = a$

on squaring both sides we get $(b\sqrt{5})^2 = (a)^2$

$\Rightarrow$ $5b^2 = a^2$

Since 5 divides a^2, then 5 divides a. (since if 'p' a prime no. divides a^2, it surely divides a).

Now let $a = 5c$ (for any positive integer c).

Substituting, $5b^2 = a^2$

$\Rightarrow$ $5b^2 = (5c)^2$

$\Rightarrow$ $5b^2 = 25c^2$

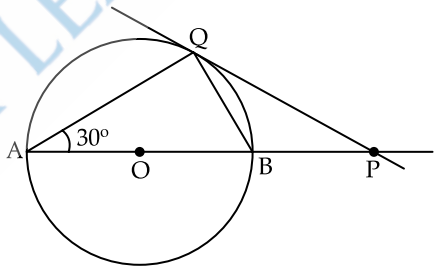
$\Rightarrow$ $b^2 = 5c^2$

Here since 5 divides b^2, we know that 5 divides b.

However this means that 'a' and 'b' have a common factor 5, apart from 1. This contradicts the fact that they are co-primes. This contradiction arose due to incorrect assumption ie, that $\sqrt{5}$ is rational.

Hence, we conclude that $\sqrt{5}$ is irrational.

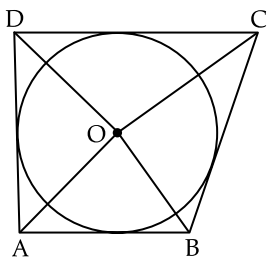
28. **(a)** In the given figure, PQ is tangent to a circle centred at O and $\angle BAQ = 30°$; show that BP = BQ. **3**



OR

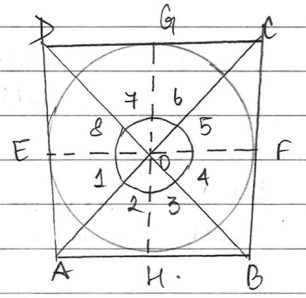
(b) In the given figure, AB, BC, CD and DA are tangents to the circle with centre O forming a quadrilateral ABCD. **3**

Show that $\angle AOB + \angle COD = 180°$



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Ans. 16) Given : AB, BC, CD and DA are tangents to circle with centre O, forming quad. ABCD



To prove : $\angle AOB + \angle COD = 180°$

$\Rightarrow \angle 2 + \angle 3 + \angle 6 + \angle 7 = 180°$

Construction : Draw 2 diameters ie, EF and GH.
Join OG, OF, OH and OE.

Proof : We know that tangents drawn from an external point to a circle are equally inclined to the centre of the circle ie,

$\angle 1 = \angle 2$; $\angle 3 = \angle 4$; $\angle 5 = \angle 6$; $\angle 7 = \angle 8$ → (1)
(since AB, BC, CD and DA are tangents to same circle).

Also, sum of all angles around a point is 360°. So,

$\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 = 360°$

from (1), we get $2(\angle 2) + 2(\angle 3) + 2(\angle 6) + 2(\angle 7) = 360°$

$2[\angle 2 + \angle 3 + \angle 6 + \angle 7] = 360°$

$\angle 2 + \angle 3 + \angle 6 + \angle 7 = 180°$

$\angle AOB + \angle COD = 180°$

Hence proved that $\angle AOB + \angle COD = 180°$.

29. Prove that $\dfrac{1 + \sec\theta - \tan\theta}{1 + \sec\theta + \tan\theta} = \dfrac{1 - \sin\theta}{\cos\theta}$ **3**

Ans. 29. To prove : $\dfrac{1 + \sec\theta - \tan\theta}{1 + \sec\theta + \tan\theta} = \dfrac{1 - \sin\theta}{\cos\theta}$

Proof : Consider LHS.

$\Rightarrow \dfrac{(\sec\theta - \tan\theta + 1)}{(\tan\theta + \sec\theta + 1)}$

$\Rightarrow \dfrac{\sec\theta - \tan\theta + (\sec^2\theta - \tan^2\theta)}{(\tan\theta + \sec\theta + 1)}$ $\quad [\because 1 + \tan^2\theta = \sec^2\theta]$

$\Rightarrow \dfrac{(\sec\theta - \tan\theta)[1 + (\sec\theta + \tan\theta)]}{(\tan\theta + \sec\theta + 1)}$ $\quad [\because (a^2 - b^2) = (a - b)(a + b)]$

$\Rightarrow \dfrac{(\sec\theta - \tan\theta)(1 + \sec\theta + \tan\theta)}{(\tan\theta + \sec\theta + 1)}$

$\Rightarrow \sec\theta - \tan\theta$.

$\Rightarrow \dfrac{1}{\cos\theta} - \dfrac{\sin\theta}{\cos\theta} = \dfrac{1 - \sin\theta}{\cos\theta}$ ie, RHS.

thus LHS = RHS. Hence proved.

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30. In a test, the marks obtained by 100 students (out of 50) are given below : **3**

Marks obtained :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of students :	12	23	34	25	6

Find the mean marks of the students.

Ans.

Let assumed mean (A) for the data = 25

class interval (marks)	x_i [upper limit + lower limit / 2]	f_i (frequency)	d_i $(x_i - A)$	$f_i d_i$
0 – 10	5	12	(5 – 25) ⇒ –20	–240
10 – 20	15	23	(15 – 25) ⇒ –10	–230
20 – 30	A 25	34	(25 – 25) ⇒ 0	0
30 – 40	35	25	(35 – 25) ⇒ 10	250
40 – 50	45	6	(45 – 25) ⇒ 20	120
Total		100		(–100)

Now using the 'Assumed mean method', mean $= \left(A + \frac{\Sigma f_i d_i}{\Sigma f_i}\right)$

substituting the values, $25 + \left[\frac{-100}{100}\right]$

$\Rightarrow 25 - 1$

$\Rightarrow 24$

Thus the mean marks obtained by the students is 24.

31. In a 2-digit number, the digit at the unit's place is 5 less than the digit at the ten's place. The product of the digits is 36. Find the number. **3**

Ans.

Let the digit be in the form $(10x + y)$. So, digit in units place $= y$ and digit in tens place $= x$.

Now, we know $y = (x - 5) \rightarrow$ (1) [given]

also product of digits $= 36$.

$xy = 36$

$x(x - 5) = 36$ [substituting value of y from (1)]

$x^2 - 5x = 36$

$x^2 - 5x - 36 = 0$

by splitting the middle term we get,

$x^2 - 9x + 4x - 36 = 0$

$x(x - 9) + 4(x - 9) = 0$

$(x - 9)(x + 4)$

$\therefore x = 9$ or $x = -4$

We reject $(x = -4)$ as this number cannot be negative.

So, $x = 9$. from (1), we get, value of $y \Rightarrow (9 - 5) = 4$

Thus the number is $10x + y \Rightarrow 94$.

Hence the required number is 94.

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SECTION - D

This section consists of 4 questions of 5 marks each.

32. (a) Using graphical method, solve the following system of equations : **5**

$3x + y + 4 = 0$ and $3x - y + 2 = 0$

Ans. (a) Consider the given equations.

(i) $3x + y + 4 = 0$

Required values of x and y are :

x	1	0	(−3)
y	(−7)	(−4)	5

(ii) $3x - y + 2 = 0$

Required values of x and y are :

x	1	0	(−2)
y	5	2	(−4)

[Lines are made on graph on page 23]

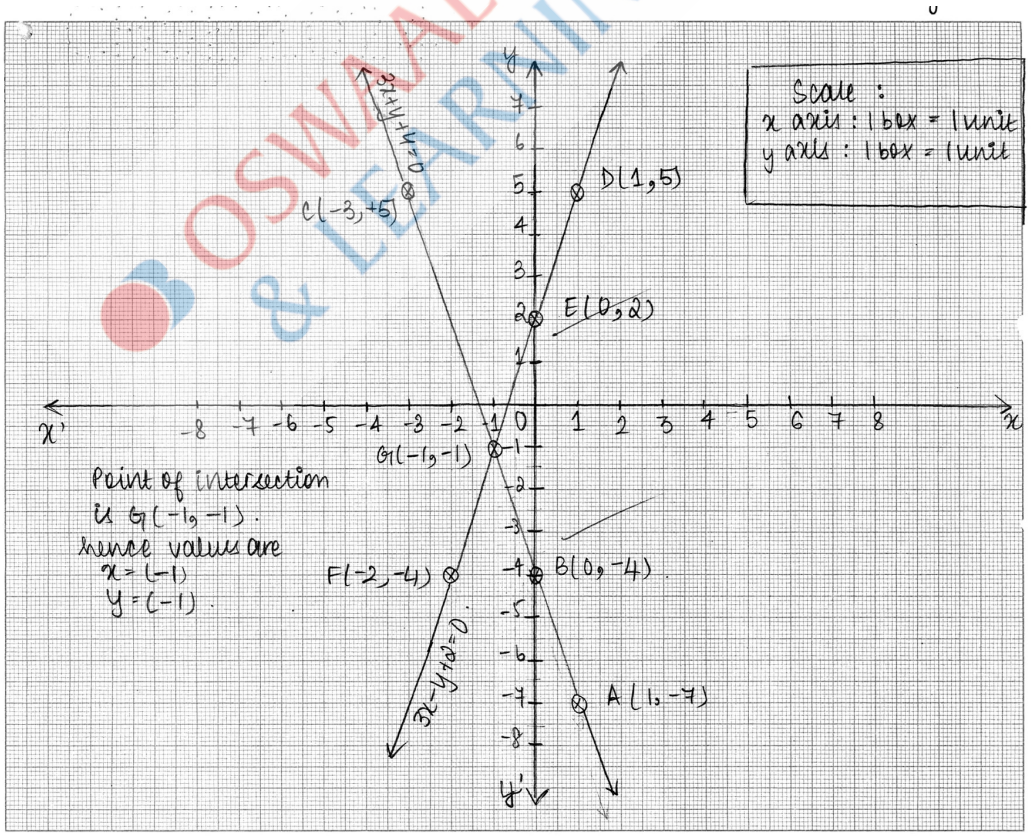
From page 23,

Point of intersection is G(−1, −1).

Hence value of $x = -1$ and $y = -1$.

We observe that this value satisfies each equation.

Hence solved.



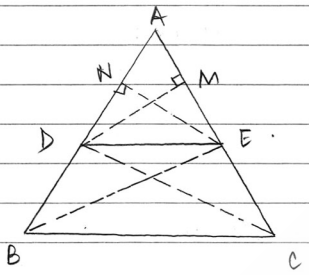
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OR

(b) Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test. **5**

33. (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio. **5**

Ans. (a) **Statement:** If a line is drawn parallel to one side of a triangle to intersect the other 2 sides at two distinct points, the other two sides are divided in the same ratio.



Given: $\triangle ABC$ where $DE \parallel BC$ and DE intersects AB at D and intersects AC at E.

To prove: $\frac{AB}{BD} = \frac{AE}{EC}$

Construction: Draw $DM \perp AC$ and $EN \perp AB$. Join BE and AC.

Proof: Consider $\triangle ADE$ and $\triangle BDE$.

area $\triangle ADE$ = 1/2 × base × height ⇒ 1/2 × AD × EN

area $\triangle BDE$ = 1/2 × base × height → 1/2 × BD × EN

$$\rightarrow \frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times BD \times EN} = \boxed{\frac{AD}{BD}} \quad \text{— (1)}$$

Similarly, consider $\triangle ADE$ and $\triangle CDE$.

area $\triangle ADE$ = 1/2 × base × height ⇒ 1/2 × AE × DM

area $\triangle CDE$ = 1/2 × base × height ⇒ 1/2 × EC × DM

$$\rightarrow \frac{ar(\triangle ADE)}{ar(\triangle CDE)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \boxed{\frac{AE}{EC}} \quad \text{— (2)}$$

Since $\triangle BDE$ and $\triangle CDE$ lie on the same base DE and between the same parallel lines (BC and DE), their area is equal. Hence (1) and (2) are equal.

from (1) and (2), $\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$

Hence proved.

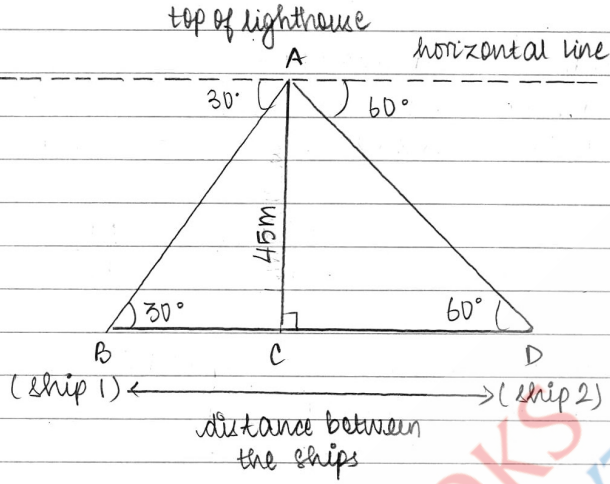
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OR

(b) Sides AB and AC and median AD to ΔABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that ΔABC ~ ΔPQR. **5**

34. From the top of a 45 m high light house, the angles of depression of two ships, on the opposite side of it, are observed to be 30° and 60°. If the line joining the ships passes through the foot of the light house, find the distance between the ships. (Use $\sqrt{3} = 1.73$) **5**

Ans.



To find : Distance between ships ie, BD.

Solution : Consider △ ABC where ∠B = 30° and ∠C = 90°

$$\tan A = \frac{AC}{BC}$$

$$\Rightarrow \tan 30° = \frac{45m}{BC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{45}{BC}$$

$$\therefore \text{Length of } BC = 45\sqrt{3}m \quad — (1)$$

now consider △ACD where ∠D = 60° and ∠C = 90°.

$$\tan 60° = \frac{AC}{CD}$$

$$\sqrt{3} = \frac{45}{CD}$$

$$CD = \frac{45}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{45\sqrt{3}}{3}$$

$$= 15\sqrt{3}m$$

so length of $CD = 15\sqrt{3}m$ — (2)

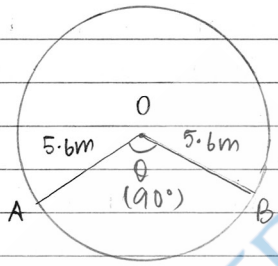
4 1
17.3
X 6
103.8

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now length BD = BC + CD
= $45\sqrt{3} + 15\sqrt{3}$m (from ① and ②)
= $\sqrt{3}[45+15]$
= (60)(1·73) = 103·8m
Thus distance between 2 ships is 103·8m or $\sqrt{3}(60)$m.

35. The perimeter of a certain sector of a circle of radius 5.6 m is 20.0 m. Find the area of the sector. **5**

Ans.



Given: circle with centre O and radius 5·6m.
Perimeter of sector OAB is 20m
To find: Area of sector OAB.
Solution: Consider sector OAB.
Perimeter of sector = 20m
2 radius + arc = 20m
2(5·6) + arc = 20
length of arc = 8·8m

length of arc = $\frac{\theta}{360^\circ} \times 2\pi r$.

$\Rightarrow \frac{\theta}{360^\circ} \times 2 \times \frac{22}{7} \times 5{\cdot}6 = 8{\cdot}8$

$$\theta = \frac{8{\cdot}8 \times 360 \times 7}{2 \times 22 \times 5{\cdot}6}$$

$$= \frac{4 \times 18}{0{\cdot}8} = 90^\circ$$

Hence the required angle (θ) = 90°
Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

Substituting values, $\frac{90}{4\ 360} \times \frac{22}{7} \times 5{\cdot}6 \times 5{\cdot}6\ 0{\cdot}8$

$\Rightarrow \frac{22 \times 5{\cdot}6 \times 8}{10\ 5}$

$\Rightarrow \frac{123{\cdot}2}{5}$

$\Rightarrow 24{\cdot}64 m^2$

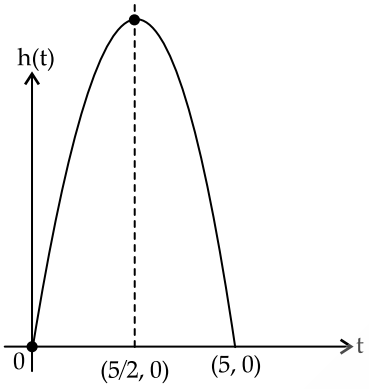
Thus the required area of the sector is $24{\cdot}64 m^2$

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SECTION - E

This section consists of 3 case based questions of 4 marks each.

36. A ball is thrown in the air so that t seconds after it is thrown, its height h metre above its starting point is given by the polynomial $h = 25t - 5t^2$.



Observe the graph of the polynomial and answer the following questions :

(i) Write zeroes of the given polynomial. **1**

(ii) Find the maximum height achieved by ball. **1**

(iii) (a) After throwing upward, how much time did the ball take to reach to the height of 30 m ? **2**

OR

(iii) (b) Find the two different values of t when the height of the ball was 20 m. **2**

Ans. 36. (i) Zeroes of the polynomial are given by the x-coordinate at the point of intersection with the coordinate axes.
Hence, zeroes of polynomial are 5 and 0.

(ii) The maximum height was achieved by the ball after 5/2 seconds.

$h = 25t - 5t^2$.

$= \left(25 \times \frac{5}{2}\right) - \left(5 \times \frac{25}{4}\right)$.

$\Rightarrow \frac{125}{2} - \frac{125}{4}$.

$\Rightarrow \frac{250 - 125}{4}$.

$\Rightarrow \frac{125}{4}$

$\Rightarrow 31.25\text{m}$

hence maximum height was 31.25m

(iii) (b) $h = 25t - 5t^2$

$20 = 25t - 5t^2$.

$0 = 25t - 5t^2 - 20$.

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$5t^2 - 25t + 20 = 0$

$t^2 - 5t + 4 = 0$

$t^2 - t - 4t + 4 = 0$

$t(t-1) - 4(t-1) = 0$

$(t-1)(t-4)$

Thus the two different values of t are '1' and '4'.

37. The word 'circus' has the same root as 'circle'. In a closed circular area, various entertainment acts including human skill and animal training are presented before the crowd.

A circus tent is cylindrical upto a height of 8 m and conical above it. The diameter of the base is 28 m and total height of tent is 18.5 m.



Based on the above, answer the following questions :

(i) Find slant height of the conical part. **1**

(ii) Determine the floor area of the tent. **1**

(iii) (a) Find area of the cloth used for making tent. **2**

OR

(iii) (b) Find total volume of air inside an empty tent. **2**

Ans. 37. (i) $l = \sqrt{h^2 + r^2}$

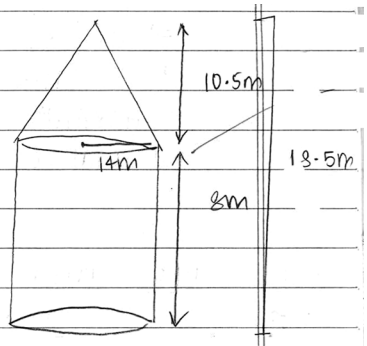
$= \sqrt{(10.5)^2 + (14)^2}$

$= \sqrt{110.25 + 196}$

$= \sqrt{306.25}$

$= 17.5\text{m}$

Hence the slant height is 17.5m



(ii) Floor area $= \pi r^2$

$= \pi(14)^2$

$= \frac{22}{7} \times 14 \times 14$

$= 44 \times 14$

$= 616\text{m}^2$

Thus floor area is 616m^2

(iii) (b) Volume = volume of cylinder + volume of cone

$= \pi r^2 h + 1/3 \pi r^2 h$

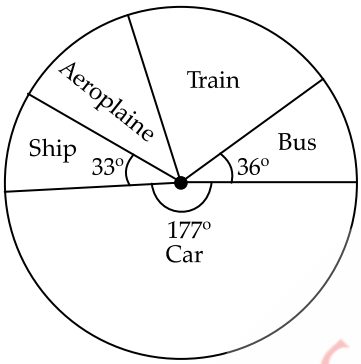
$= \pi r^2 (h + 1/3 h)$

$= \left[\frac{22}{7} \times 14 \times 14\right]\left[8 + \left(\frac{1}{3} \times 10.5\right)\right]$

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$\Rightarrow 616(8+3.5)$
$\Rightarrow 616(11.5)$
$\Rightarrow 7084m^3$
Thus total volume is $7084m^3$

38. In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey.



Observe the pie chart and answer the following questions :

(i) If one person is selected at random, find the probability that he/she travelled by bus or ship. **1**

(ii) Which is most favourite mode of transport and how many people used it ? **1**

(iii) (a) A person is selected at random. If the probability that he did not use train is $\frac{4}{5}$, find the number of people who used train. **2**

OR

(iii) (b) The probability that randomly selected person used aeroplane is $\frac{7}{60}$. Find the revenue collected by air company at the rate of ₹ 5,000 per person. **2**

Ans. 38. (i) P(E) = Probability of bus or ship
$= \frac{\text{no. of outcomes favourable to E}}{\text{total possible outcomes of E}}$
$= \boxed{\frac{23}{120}}$

since no. of people who travel by bus or ship $= \frac{69}{360} \times 120 \Rightarrow 23$

Required probability is $\boxed{\frac{23}{120}}$

(ii) 'Car' is the favourite mode of transport.
No. of people who travel by it $= \frac{177}{360} \times 120$
$= 59$ people
Out of the 120 surveyed people, 59 use cars.

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(iii) (b) No. of people who use planes = $\frac{7}{60} \times 120 \Rightarrow$ 14 people.

Thus, total revenue = ₹ (5000)(14)

= ₹ 70,000

Thus total money made is ₹ 70,000

■■

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Mathematics Standard
CBSE Examination Paper – 2023

With Topper's Answer

Time : 3:00 Hours **Max. Marks : 80**

General Instructions:

Read the following instructions carefully and follow them:

(i) *This question paper contains **38** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **FIVE** sections - **A, B, C, D** and **E.***

(iii) *In **Section A,** Question numbers **1** to **18** are multiple choice questions (MCQs) and question numbers **19** and **20** are Assertion – Reason based questions of **1** mark each.*

(iv) *In **Section B,** Question numbers **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C,** Question numbers **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D,** Question numbers **32** to **35** are long answer (LA) type questions, carrying **5** marks each.*

(vii) *In **Section E,** Question numbers **36** to **38** are **case-study based integrated** questions carrying **4** marks each. Internal choice is provided in **2** marks questions in each case-study.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section **B**, 2 questions in Section **C**, 2 questions in Section **D** and 3 questions of 2 marks in Section **E**.*

(ix) *Draw neat diagrams wherever required. Take $\pi = \frac{22}{7}$ wherever required, if not stated.*

(x) *Use of calculators is **NOT allowed**.*

Set--30/5/1

SECTION – A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. The number of polynomials having zeroes -3 and 5 is :

(a) only one **(b)** infinite **(c)** exactly two **(d)** at most two

Ans. (b) infinite

2. The pair of equations $ax + 2y = 9$ and $3x + by = 18$ represent parallel lines, where a, b are integers, if :

(a) $a = b$ **(b)** $3a = 2b$ **(c)** $2a = 3b$ **(d)** $ab = 6$

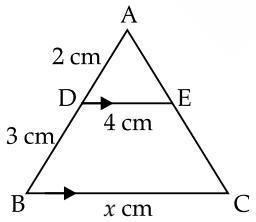
Ans. (d) $ab = 6$

3. The common difference of the A.P. whose nth term is given by $a_n = 3n + 7$, is :

(a) 7 **(b)** 3 **(c)** $3n$ **(d)** 1

Ans. (b) 3

4. In the given figure, DE || BC. The value of x is:



(a) 6 **(b)** 12.5 **(c)** 8 **(d)** 10

Ans. (d) 10

5. A quadratic equation whose roots are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$ is :

(a) $x^2 - 4x + 1 = 0$ **(b)** $x^2 + 4x + 1 = 0$ **(c)** $4x^2 - 3 = 0$ **(d)** $x^2 - 1 = 0$

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Ans. (a) $x^2 - 4x + 1 = 0$

6. If $\tan\theta = \frac{5}{12}$, then the value of $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta}$ is :

(a) $-\frac{17}{7}$ **(b)** $\frac{17}{7}$ **(c)** $\frac{17}{13}$ **(d)** $-\frac{7}{13}$

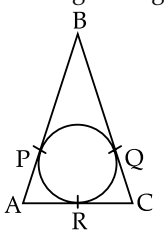
Ans. (a) $-\frac{17}{7}$

7. The distance between the points $P\left(-\frac{11}{3}, 5\right)$ and $Q\left(-\frac{2}{3}, 5\right)$ is :

(a) 6 units **(b)** 4 units **(c)** 2 units **(d)** 3 units

Ans. (d) 3 units

8. In the given figure, AB = BC = 10 cm. If AC = 7 cm, then the length of BP is :



(a) 3.5 cm **(b)** 7 cm **(c)** 6.5 cm **(d)** 5 cm

Ans. (c) 6.5 cm

9. Water in a river which is 3 m deep and 40 m wide is flowing at the rate of 2 km/h. How much water will fall into the sea in 2 minutes ?

(a) 800 m^3 **(b)** 4000 m^3 **(c)** 8000 m^3 **(d)** 2000 m^3

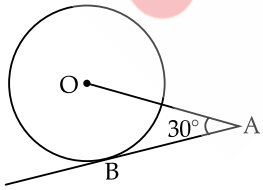
Ans. (c) 8000 m^3

10. If the mean and the median of a data are 12 and 15 respectively, then its mode is:

(a) 13.5 **(b)** 21 **(c)** 6 **(d)** 14

Ans. (b) 21

11. In the given figure, AB is a tangent to the circle centered at O. If OA = 6 cm and $\angle OAB = 30°$, then the radius of the circle is:



(a) 3 cm **(b)** $3\sqrt{3}$ cm **(c)** 2 cm **(d)** $\sqrt{3}$ cm

Ans. (a) 3 cm

12. $\left(\frac{2\tan 30°}{1+\tan^2 30°}\right)$ is equal to :

(a) sin 60° **(b)** cos 60° **(c)** tan 60° **(d)** sin 30°

Ans. (a) sin 60°

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13. In Δ ABC and Δ DEF, $\frac{AB}{DE}=\frac{BC}{FD}$. Which of the following makes the two triangles similar ?

(a) $\angle A = \angle D$ **(b)** $\angle B = \angle D$ **(c)** $\angle B = \angle E$ **(d)** $\angle A = \angle F$

Ans. (b) $\angle B = \angle D$

14. The 11th term from the end of the A.P. : 10, 7, 4, ... ,– 62 is :

(a) 25 **(b)** 16 **(c)** – 32 **(d)** 0

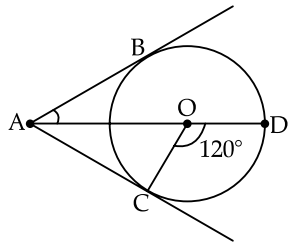
Ans. (c) – 32

15. Two coins are tossed together. The probability of getting at least one tail is :

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** 1

Ans. (c) $\frac{3}{4}$

16. In the given figure, AC and AB are tangents to a circle centered at O. If $\angle COD = 120°$, then $\angle BAO$ is equal to:



(a) 30° **(b)** 60° **(c)** 45° **(d)** 90°

Ans. (a) 30°

17. Which of the following numbers cannot be the probability of happening of an event?

(a) 0 **(b)** $\frac{7}{0.01}$ **(c)** 0.07 **(d)** $\frac{0.07}{3}$

Ans. (b) $\frac{7}{0.01}$

18. If every term of the statistical data consisting of n terms is decreased by 2, then the mean of the data :

(a) decreases by 2 **(b)** remains unchanged

(c) decreases by $2n$ **(d)** decreases by 1

Ans. (a) decreases by 2

Questions number **19** and **20** are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled as Assertion (a) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion (a) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (a).

(b) Both Assertion (a) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (a).

(c) Assertion (a) is true, but Reason (R) is false.

(d) Assertion (a) is false, but Reason (R) is true.

19. Assertion (a): If the points A(4, 3) and B(x, 5) lie on a circle with centre O(2, 3), then the value of x is 2.
Reason (R): Centre of a circle is the mid-point of each chord of the circle.

Ans (c) Assertion (A) is true, but Reason (R) is false.

20. Assertion (a): The number 5^n cannot end with the digit 0, where n is a natural number.
Reason (R): Prime factorisation of 5 has only two factors, 1 and 5.

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Ans. (c) Assertion (A) is true, but Reason (R) is false.
[only 1 prime factor = 5 ∵ prime factorisation of a prime number is the number itself]

SECTION – B

This section comprises very short answer (VSA) type questions of 2 marks each.

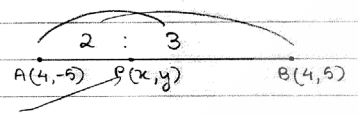
21. (a) The line segment joining the points A(4,–5) and B(4, 5) is divided by the point P such that AP : AB = 2 : 5. Find the coordinates of P.

OR

(b) Point P(x, y) is equidistant from points A(5, 1) and B(1, 5). Prove that $x = y$.

Ans. A(4, −5) and B(4, 5)

(a) Let the coordinates of point P which divides AB such that AP : AB = 2 : 5 be P(x, y).



$\frac{AP}{AB} = \frac{2}{5}$

$\Rightarrow \frac{AP + PB}{AP} = \frac{5}{2}$

$\Rightarrow 2PB = 3AP \Rightarrow AP : PB = 2 : 3$

P divides AB internally in the ratio of 2 : 3

By section formula,

$P(x, y) = P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

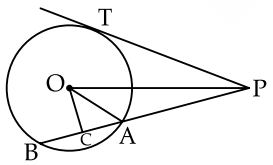
$\Rightarrow P(x, y) = P\left(\frac{2\times4 + 3\times4}{2+3}, \frac{2\times5 + 3\times(-5)}{2+3}\right)$

$\Rightarrow P(x, y) = P\left(\frac{20}{5}, \frac{-5}{5}\right) = P(4, -1)$

∴ Coordinates of P are P(4, −1).

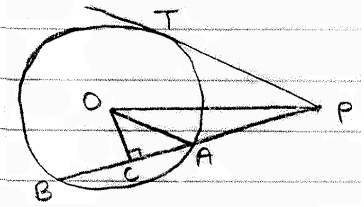
22. In the given figure, PT is a tangent to the circle centered at O. OC is perpendicular to chord AB. Prove that PA.

$PB = PC^2 - AC^2$.



Ans. Given: Circle with centre O
PT is a tangent
OC ⊥ AB

To Prove: $PA \cdot PB = PC^2 - AC^2$



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Perpendicular from centre to chord bisects it
$\Rightarrow$ C is midpoint of AB ($\because \angle OCA = 90°$, given)
$\Rightarrow BC = CA = \frac{AB}{2}$

$\Rightarrow AB = 2AC$

By Pythagoras Theorem,
In right $\triangle OCA$, $OA^2 = OC^2 + AC^2$
In right $\triangle OCP$, $OP^2 = OC^2 + PC^2$

LHS = PA · PB
= PA × (PA + AB)
= $PA^2 + PA \cdot AB$
= $PA^2 + PA \cdot 2AC$ (from ①)
= $(PA)^2 + 2(PA)(AC)$
= $(PA + AC)^2 - AC^2$ [$(a+b)^2 = a^2 + 2ab + b^2$]
= $PC^2 - AC^2$ = RHS
Hence, proved

23. Using prime factorisation, find HCF and LCM of 96 and 120.

Ans. First number = 96
Second number = 120

$96 = 2^5 \times 3$
$120 = 2^3 \times 3 \times 5$
HCF (96, 120) = $2^3 \times 3$
= 8 × 3
= 24

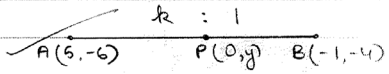
2 | 96	2 | 120
2 | 48	2 | 60
2 | 24	2 | 30
2 | 12	3 | 15
2 | 6	5
3	

LCM (96, 120) = $2^5 \times 3 \times 5$
= 32 × 3 × 5
= 480

$\therefore$ HCF (96, 120) = 24 and LCM (96, 120) = 480

24. Find the ratio in which y-axis divides the line segment joining the points (5, – 6) and (– 1, – 4).

Ans. Points are A(5, -6) and B(-1, -4)
Let the point where y-axis ($x = 0$) intersects AB be P(0, y)



Let the ratio in which P(0, y) divides AB be k : 1

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By section formula,

$P(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

$P(0, y) = \left(\frac{-k+5}{k+1}, \frac{-4k+(-6)}{k+1}\right)$

$0 = \frac{-k+5}{k+1}$ and $y = \frac{-4k-6}{k+1}$

$0 = -k + 5$

$k = 5$

$k : 1 = 5 : 1$

∴ Ratio is $\boxed{5:1}$

25. (a) If $a \cos\theta + b \sin\theta = m$ and $a \sin\theta - b \cos\theta = n$, then prove that $a^2 + b^2 = m^2 + n^2$.

OR

(b) Prove that :

$$\sqrt{\frac{\sec A - 1}{\sec A + 1}} + \sqrt{\frac{\sec A + 1}{\sec A - 1}} = 2\operatorname{cosec} A$$

Ans.

(a) $a\cos\theta + b\sin\theta = m$

$a\sin\theta - b\cos\theta = n$

To prove: $a^2 + b^2 = m^2 + n^2$

$a\cos\theta + b\sin\theta = m$

Squaring both sides,

$a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta = m^2$ $\left[(a+b)^2 = a^2 + b^2 + 2ab\right]$

$a\sin\theta - b\cos\theta = n$

Squaring both sides,

$a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta = n^2$ $\left[(a-b)^2 = a^2 + b^2 - 2ab\right]$

Adding ① and ②,

$(a^2\cos^2\theta + a^2\sin^2\theta) + (b^2\sin^2\theta + b^2\cos^2\theta) + 2ab\sin\theta\cos\theta - 2ab\sin\theta\cos\theta = m^2 + n^2$

$\Rightarrow a^2(\sin^2\theta + \cos^2\theta) + b^2(\sin^2\theta + \cos^2\theta) = m^2 + n^2$

$\because \sin^2\theta + \cos^2\theta = 1$

$\Rightarrow a^2 + b^2 = m^2 + n^2$

LHS = RHS

Hence, proved

SECTION – C

This section comprises short answer (SA) type questions of 3 marks each.

26. (a) Prove that $\sqrt{3}$ is an irrational number.

OR

(b) The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they change simultaneously at 7 a.m., at what time will they change together next ?

Ans.

(a) Let us assume, to the contrary, that $\sqrt{3}$ is rational.

$\Rightarrow \sqrt{3} = \frac{p}{q}$ where $q \neq 0$, p and q are coprime positive integers

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Squaring both sides,

$\Rightarrow 3 = \frac{p^2}{q^2}$

$\Rightarrow p^2 = 3q^2$

$\Rightarrow$ 3 divides p^2

$\Rightarrow$ 3 divides p ($\because$ 3 is prime)

So, let $p = 3m$

Substituting in

$(3m)^2 = 3q^2$

$\Rightarrow 9m^2 = 3q^2$

$\Rightarrow q^2 = 3m^2$

$\Rightarrow$ 3 divides q^2

$\Rightarrow$ 3 divides q

From ③ and ④,

3 divides both p and q

But p and q are coprime, i.e. HCF$(p, q) = 1$ (Using ①)

which is a contradiction

$\therefore$ our supposition is wrong

$\therefore \sqrt{3}$ must be irrational.

Hence, proved

27. If p^{th} term of an A.P. is q and q^{th} term is p, then prove that its nth term is $(p + q - n)$.

Ans. Let the first term of AP be a

and common difference be d

AP: $a, a+d, a+2d, \ldots$

$a_n = a + (n-1)d$

$a_p = a + (p-1)d = q$

$\Rightarrow a + pd - d = q$ ①

$a_q = a + (q-1)d = p$

$\Rightarrow a + qd - d = p$ ②

Subtracting ① from ②,

$a + qd - d = p$

$\ominus \quad a + pd - d = q$

$(q-p)d = p-q$

$\Rightarrow d = -1$

Substituting in ①,

$a + p(-1) - (-1) = q$

$\Rightarrow a + 1 = p + q$

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$a_n = a + (n-1)d$

$= a + (n-1)(-1)$

$= a - n + 1$

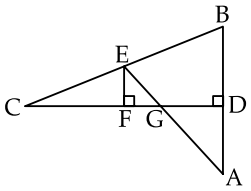
$= (a+1) - n$

$= p + q - n$ (Substituting from (3))

LHS = RHS

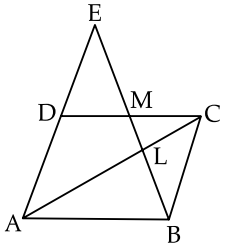
Hence, proved

28. (a) In the given figure, CD is the perpendicular bisector of AB. EF is perpendicular to CD. AE intersects CD at G. Prove that $\frac{CF}{CD} = \frac{FG}{DG}$.



OR

(b) In the given figure, ABCD is a parallelogram. BE bisects CD at M and intersects AC at L. Prove that EL = 2BL.

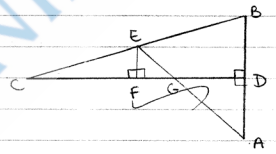


Ans.

(a) Given: CD is ⊥ bisector of AB

EF ⊥ CD

AE intersects CD at G



To Prove: $\frac{CF}{CD} = \frac{FG}{DG}$

Proof: In △EFG and △ADG,

∠EFG = ∠ADG = 90°	(given, linear pair with ∠EFC and ∠BDC)
∠EGF = ∠AGD	(vertically opposite angles)
∴ △EFG ~ △ADG	(by AA similarity criterion)
$\Rightarrow \frac{EF}{AD} = \frac{FG}{DG}$	(cpst)

In △ECF and △BCD,

∠EFC = ∠BDC = 90°	(given)
∠ECF = ∠BCD	(common angle)
∴ △ECF ~ △BCD	(by AA similarity criterion)
$\Rightarrow \frac{EF}{BD} = \frac{CF}{CD}$	(cpst)

But AD = BD (∵ CD bisects AB)

$\Rightarrow \frac{EF}{AD} = \frac{EF}{BD} = \frac{CF}{CD}$

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From item ① and ②,

$$\frac{EF}{AD} = \frac{CF}{CD} = \frac{FG}{DG}$$

$$\therefore \frac{FG}{CD}\,\frac{CF}{DG} = \frac{FG}{DG}$$

Hence, proved

29. Two people are 16 km apart on a straight road. They start walking at the same time. If they walk towards each other with different speeds, they will meet in 2 hours. Had they walked in the same direction with same speeds as before, they would have met in 8 hours. Find their walking speeds.

Ans. Let the speed of person 1 be x km/h and speed of person 2 be y km/h $(x > y)$

distance = speed × time

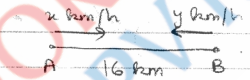
Given distance = 16 km

Case 1 : Towards each other

Time = 2 h

Speed = $(x+y)$ km/h

Distance = $2(x+y)$ km



$\Rightarrow 16 = 2x + 2y$

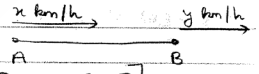
$\Rightarrow x + y = 8$ ... ①

Case 2 Same direction

Time = 8 h

Speed = $(x-y)$ km/h

Distance (only AB) = $8(x-y)$



$[16 + 8y = 8x]$

$\Rightarrow 16 = 8x - 8y$

$\Rightarrow x - y = 2$ ... ②

Adding ① and ②,

$x + y = 8$

$+ \quad x - y = 2$

$2x = 10$

$\Rightarrow x = 5$

Substituting in ①,

$y = 3$ km

$\therefore$ Speed of person 1 = 5 km/h

Speed of person 2 = 3 km/h

30. Prove that :

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\mathrm{cosec}\,\theta$$

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Ans. $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$

$$\text{LHS} = \frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta}$$

$$= \frac{\tan\theta}{1-\frac{1}{\tan\theta}} + \frac{\frac{1}{\tan\theta}}{1-\tan\theta}$$

$$= \frac{\tan^2\theta}{\tan\theta - 1} + \frac{1}{\tan\theta(1-\tan\theta)}$$

$$= \frac{\tan^2\theta}{\tan\theta - 1} - \frac{1}{\tan\theta(\tan\theta - 1)}$$

$$= \frac{\tan^3\theta - 1}{(\tan\theta - 1)\tan\theta}$$

$$= \frac{(\tan\theta - 1)(\tan^2\theta + 1 + \tan\theta)}{\tan\theta(\tan\theta - 1)}$$

$\left[\frac{(a-b)(a^2+b^2+ab)}{a^3-b^3}\right]$

$$= \frac{1 + \tan^2\theta + \tan\theta}{\tan\theta}$$

$$= \frac{\tan\theta + \sec^2\theta}{\tan\theta}$$

$(1 + \tan^2\theta = \sec^2\theta)$

$$= \frac{\tan\theta}{\tan\theta} + \frac{\sec^2\theta}{\tan\theta}$$

$$= 1 + \frac{1}{\cos^2\theta} \times \frac{\cos\theta}{\sin\theta}$$

$\left(\sec\theta = \frac{1}{\cos\theta},\ \tan\theta = \frac{\sin\theta}{\cos\theta}\right)$

$$= 1 + \frac{1}{\cos\theta \sin\theta}$$

$$= 1 + \sec\theta\,\text{cosec}\,\theta$$

$$= \text{RHS}$$

Hence, proved

31. Find the mean of the following frequency distribution:

Classes	25-30	30-35	35-40	40-45	45-50	50-55	55-60
Frequency	14	22	16	6	5	3	4

Ans.

Classes	Frequency (f_i)	Midpoints (x_i)	$u_i = \frac{x_i - A}{h}$	$f_i u_i$
25 - 30	14	27.5	-3	-42
30 - 35	22	32.5	-2	-44
35 - 40	16	37.5	-1	-16
40 - 45	6	A (42.5)	0	0
45 - 50	5	47.5	1	5
50 - 55	3	52.5	2	6
55 - 60	4	57.5	3	12
	$\Sigma f_i = 70$			$\Sigma f_i u_i = -79$

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Mean = $\bar{x} = A + \frac{h\Sigma f_i u_i}{\Sigma f_i}$

$= 42.5 + \frac{5 \times (-79)}{70}$

$= 42.5 - \frac{395}{70}$

$= 42.5 - 5.642$

$= 36.858 \approx 36.86$

∴ Mean = 36.86 (approx)

SECTION – D

This section comprises long answer (LA) type questions of 5 marks each.

32. One observer estimates the angle of elevation to the basket of a hot air balloon to be 60°, while another observer 100m away estimates the angle of elevation to be 30°. Find :

(a) The height of the basket from the ground.

(b) The distance of the basket from the first observer's eye.

(c) The horizontal distance of the second observer from the basket.

Ans.

Introduction

2 cases are formed

A = hot-air ballon

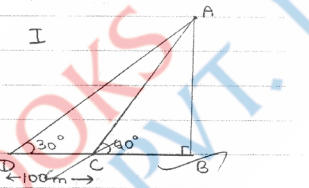
C = first observer

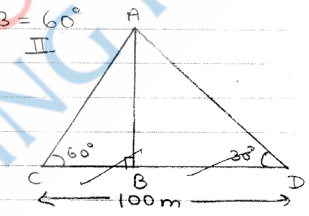
D = second observer

Angle of elevation of A from C = ∠ACB = 60°

Angle of elevation of A from BD = ∠ADB = 30°

CD = 100 m





(a) Let height of basket = AB = h m

Case I : In △ABC,

$\tan 60° = \frac{AB}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{BC}$

$\Rightarrow BC = \frac{h}{\sqrt{3}}$ (1)

In △ABD,

$\tan 30° = \frac{AB}{BD}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BC + 100}$

$\Rightarrow BC + 100 = h\sqrt{3}$

$BC + 100 = h\sqrt{3}$

$\Rightarrow \frac{h}{\sqrt{3}} + 100 = h\sqrt{3}$

$\Rightarrow 100 = h\sqrt{3} - \frac{h\sqrt{3}}{3}$

$\Rightarrow \frac{2h\sqrt{3}}{3} = 100$

$\Rightarrow h = \frac{150}{\sqrt{3}} = 50\sqrt{3}$ m

∴ Height of basket = $\boxed{50\sqrt{3} \text{ m}}$

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Case 2 : In $\triangle ABC$,

$\tan 60^\circ = \frac{AB}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{BC}$

$\Rightarrow BC = \frac{h}{\sqrt{3}}$

In $\triangle ABD$,

$\tan 30^\circ = \frac{AB}{BD}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BD}$

$\Rightarrow BD = h\sqrt{3}$

$BC + BD = 100\text{ m}$

$\Rightarrow \frac{h}{\sqrt{3}} + h\sqrt{3} = 100$

$\Rightarrow \frac{4h\sqrt{3}}{3} = 100$

$\Rightarrow h = 25\sqrt{3}$

$\therefore$ Height of basket = $\boxed{25\sqrt{3}\text{ m}}$

(b). Case 1 : Dist of A from C = AC

In $\triangle ABC$,

$\sin 60^\circ = \frac{AB}{AC}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{50\sqrt{3}}{AC}$

$\Rightarrow AC = 100\text{ m}$

$\therefore$ Distance of basket from first observer = $\boxed{100\text{ m}}$

Case 2 : Dist. of A from C = AC

In $\triangle ABC$,

$\sin 60^\circ = \frac{AB}{AC}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{25\sqrt{3}}{AC}$

$\Rightarrow AC = 50\text{ m}$

$\therefore$ Distance of basket from first observer = $\boxed{50\text{ m}}$

(c) Case 1 : To find - BD

$BD = BC + CD$

$= \frac{h}{\sqrt{3}} + 100$ (from (1))

$= \frac{50\sqrt{3}}{\sqrt{3}} + 100$

$= 50 + 100 = 150$

$\therefore$ Horizontal distance BD = $\boxed{150\text{ m}}$

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Case 2 : To find – BD

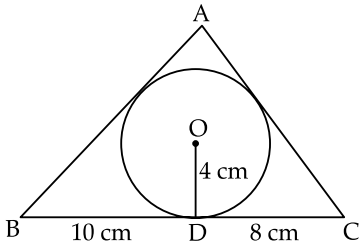
$BD = h\sqrt{3}$ (from (2))

$= 25\sqrt{3} \times \sqrt{3}$

$= 75$

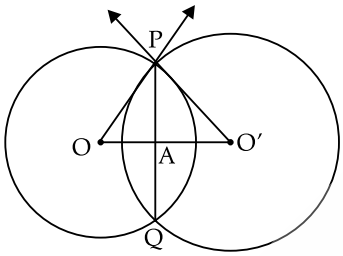
∴ Horizontal distance BD = 75 m

33. (a) A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC are of lengths 10 cm and 8 cm respectively. Find the lengths of the sides AB and AC, if it is given that area Δ ABC = 90 cm^2.



OR

(b) Two circles with centres O and O' of radii 6 cm and 8 cm, respectively intersect at two points P and Q such that OP and O'P are tangents to the two circles. Find the length of the common chord PQ.



Ans.

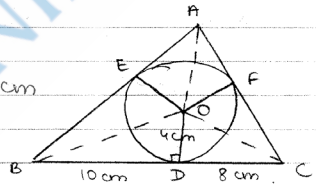
(a) Given : △ABC

Incircle with radius $r = 4$ cm

BD = 10 cm

CD = 8 cm

ar (ABC) = 90 cm^2



To find : Lengths of AB and AC.

Construction : Join OE, OF, OA, OB, OC

OD = OE = OF = r = 4cm (radii)

Tangents from same external point are equal in length.

From pt. A, AE = AF = x (let it be)

From pt. B, BE = BD = 10 cm

From pt. C, CD = CF = 8 cm

Area of $\Delta = \frac{1}{2} \times b \times h$

Area of △ABC = Area of △OAB + Area of △OBC + Area of △OCA

$\Rightarrow 90 = \frac{1}{2} \times OE \times AB + \frac{1}{2} \times OD \times BC + \frac{1}{2} \times OF \times AC$

$$+\frac{1}{2}\times 2x\times(CF+AF)$$

$$\Rightarrow \quad 90=\frac{1}{2}\times 2x(x+10)+\frac{1}{2}\times 2x\times(10+8)+\frac{1}{2}\times 2x(8+x)$$

$$\Rightarrow \quad 90=\frac{1}{2}\times 4\,(x+10+10+8+8+x)$$

$$\Rightarrow \quad 45=2x+36$$

$$\Rightarrow \quad 2x=9$$

$$\Rightarrow \quad x=4.5$$

$$AB=AE+BE=4.5+10=14.5\text{ cm}$$

$$AC=AF+CF=4.5+8=12.5\text{ cm}$$

$$\therefore \quad AB=14.5\text{ cm}$$

$$AC=12.5\text{ cm}$$

34. (a) A train travels at a certain average speed for a distance of 54 km and then travels a distance of 63 km at an average speed of 6 km/h more than the first speed. If it takes 3 hours to complete the journey, what was its first average speed ?

OR

(b) Two pipes together can fill a tank in $\frac{15}{8}$ hours. The pipe with 8 larger diameter takes 2 hours less than the pipe with smaller diameter to fill the tank separately. Find the time in which each pipe can fill the tank separately.

Ans. (a) Let the original average speed be x km/h.

Original : distance $= 54$ km

speed $= x$ km/h

$$\text{time}=\frac{\text{distance}}{\text{speed}}=\frac{54}{x}\text{ h}$$

New : distance $= 63$ km

speed $= (x+6)$ km/h

$$\text{time}=\frac{\text{distance}}{\text{speed}}=\frac{63}{x+6}\text{ h}$$

$$\frac{54}{x}+\frac{63}{x+6}=3$$

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$$\Rightarrow 3\left(\frac{18}{x}+\frac{21}{x+6}\right)=3$$

$$\Rightarrow \frac{18x+108+21x}{x^2+6x}=1$$

$$\Rightarrow 39x+108=x^2+6x$$

$$\Rightarrow x^2-33x-108=0$$

$$\Rightarrow x^2-36x+3x-108=0$$

$$\Rightarrow x(x-36)+3(x-36)=0$$

$$\Rightarrow (x-36)(x+3)=0$$

$$\Rightarrow x=36 \text{ or } x=-3$$

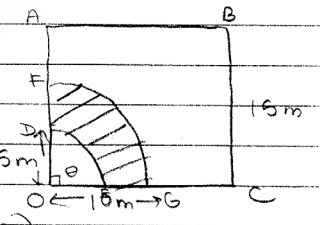
∵ speed cannot be negative,
$x=-3$ will be neglected.

$\Rightarrow x=36$

∴ Average speed of train (original) = 36 km/h

35. A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze. Also, find the increase in grazing area if length of rope is increased to 10m. (Use $\pi = 3.14$)

Ans.



Square field OABC
Side = s = 15 m

Quadrant
$\angle DOE = \theta = 90°$ (all angles of square = 90°)
Radius = length of rope
= r = 5 m

Area grazed by horse = Area of sector DOE

$$= \frac{\theta}{360^\circ} \times \pi \times r^2$$

$$= \frac{90}{360} \times \frac{157}{50} \times 5 \times 5$$

$$= \frac{3925}{200}$$

$$= 19.625 \text{ m}^2$$

157
25
785
314×
3925
2 7775
= 1962.5

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New, new radius = new length of rope = R = 10 m

Increase in grazing area = Area of sector FOG − Area of sector DOE

$$= \frac{\theta}{360^\circ} \times \pi \times R^2 - \frac{\theta}{360^\circ} \times \pi \times r^2$$

$$= \frac{90}{360} \times \frac{314}{100} \times (10^2 - 5^2)$$

$$= \frac{90}{360} \times \frac{314}{100} \times 15 \times 5 \qquad [a^2 - b^2 = (a+b)(a-b)]$$

$$= \frac{11775}{200}$$

$$= 58.875 \text{ m}^2$$

Original area grazed = 19.625 m^2

Increase in area = 58.875 m^2

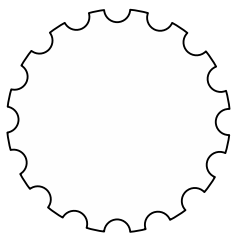
SECTION – E

This section comprises 3 case study based questions of 4 marks each.

Case Study - 1

36. A golf ball is spherical with about 300 - 500 dimples that help increase its velocity while in play. Golf balls are traditionally white but available in colours also. In the given figure, a golf ball has diameter 4.2 cm and the surface has 315 dimples (hemi-spherical) of radius 2 mm.





Based on the above, answer the following questions :

(i) Find the surface area of one such dimple. **1**

(ii) Find the volume of the material dug out to make one dimple. **1**

(iii)(a) Find the total surface area exposed to the surroundings. **2**

OR

(iii)(b) Find the volume of the golf ball. **2**

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Ans.

Golf ball = Sphere

Radius = $R = \frac{4.2}{2} = 2.1$ cm

Dimple = Hemisphere

Radius = $r = 2$ mm $= 0.2$ cm

(i) SA of 1 dimple = CSA of hemisphere

$= 2\pi r^2$

$= 2 \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10}$

$= \frac{176}{700}$ cm^2

$= 0.2514$ cm^2

(ii) Vol to make 1 dimple = Vol of hemisphere

$= \frac{2\pi r^3}{3}$

$= \frac{2}{3} \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10} \times \frac{2}{10}$

$= \frac{352}{21000}$

$= 0.01676$ cm^3

(iii) (iv) Vol of golf ball = Vol of sphere − Vol of 315 hemispheres

$= \frac{4\pi R^3}{3} - \frac{2\pi r^3 \times 315}{3}$

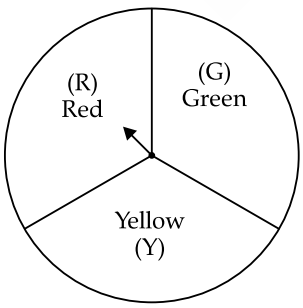
$= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{10} \times \frac{21}{10} \times \frac{21}{10} - \frac{2}{3} \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10} \times \frac{21}{10} \times 315$

$= 38.808 - 5.28$

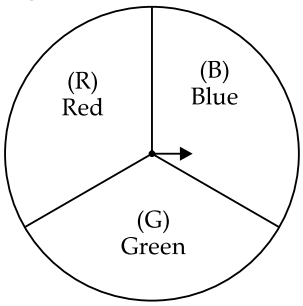
$= 33.528$ cm^3

Case Study - 2

37. A middle school decided to run the following spinner game as a fund-raiser on Christmas Carnival.



I



II

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Making Purple : Spin each spinner once. Blue and red make purple. So, if one spinner shows Red (R) and another Blue (b), then you 'win'. One such outcome is written as 'RB'.

Based on the above, answer the following questions :

(i) List all possible outcomes of the game. **1**

(ii) Find the probability of 'Making Purple'. **1**

(iii)(a) For each win, a participant gets ₹ 10, but if he/she loses, he/she has to pay ₹ 5 to the school. If 99 participants played, calculate how much fund could the school have collected. **2**

OR

(iii)(b) If the same amount of ₹ 5 has been decided for winning or losing the game, then how much fund had been collected by school? (Number of participants = 99) **2**

Ans.

(i) Spinner I – Spinner II

Red (R) – Red (R) { RR,
Red (R) – Blue (B) RB,
Red (R) – Green (G) RG,
Green (G) – Red (R) GR,
Green (G) – Blue (B) GB,
Green (G) – Green (G) GG,
Yellow (Y) – Red (R) YR,
Yellow (Y) – Blue (B) YB,
Yellow (Y) – Green (G) YG }

Total no. of outcomes = 9

(ii) X = {RB}

Favourable outcomes = 1

P(making purple) = No. of favourable outcomes / Total no. of outcomes = 1/9

(iii)(a) No. of participants = 99

Winning
No. = 1/9 × 99 = 11

Amount = ₹(−10) (school has to pay)

Value = ₹(−110)
[loss]

Loss
No. = 99 − 11 = 88

Amount = ₹5 (pay to school)

Value = ₹440
[gain]

Net = +440 − 110
= +330

∴ So, the school most likely collected Rs. 330

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Case Study - 3

38. In a pool at an aquarium, a dolphin jumps out of the water travelling at 20 cm per second. Its height above water level after t seconds is given by $h = 20t - 16t^2$.

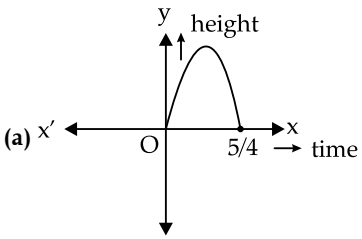


Based on the above, answer the following questions :

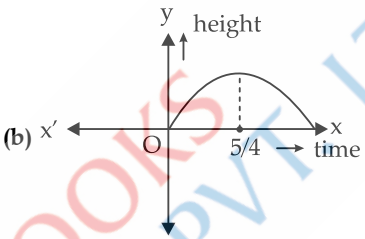
(i) Find zeroes of polynomial $p(t) = 20t - 16t^2$. **1**

(ii) Which of the following types of graph represents $p(t)$? **1**

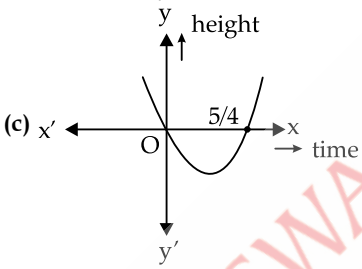
(a)



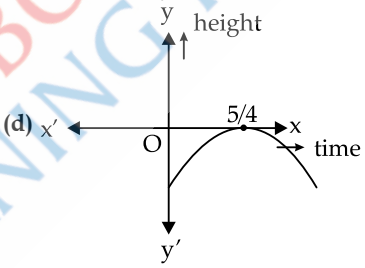
(b)



(c)



(d)



Ans.

(iii)(a) What would be the value of h at $t = \frac{3}{2}$? Interpret the result. **2**

OR

(b) How much distance has the dolphin covered before hitting the water level again ? 2

(i) $p(t) = 20t - 16t^2$

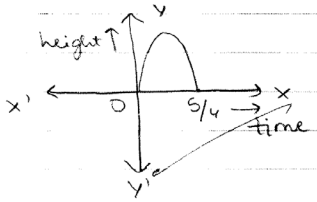
$20t - 16t^2 = 0$

$\Rightarrow -4t(4t - 5) = 0$

$t = 0$ or $t = \frac{5}{4}$

$\therefore$ Zeroes of $p(t) = 20t - 16t^2$ are 0 and $\frac{5}{4}$

(ii) (a)



→ opens downwards ($a < 0$)

→ intersects x-axis at $(0, 0)$ and $\left(\frac{5}{4}, 0\right)$

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(iii) Water level is hit when $h = 0$

(b) $h = 20t - 16t^2$

$\Rightarrow 0 = -4t(4t - 5)$

$\Rightarrow t = 0$ or $t = \frac{5}{4}$

$\because$ Delphin has started at $t = 0$

$\Rightarrow$ at $t = \frac{5}{4}$, delphin reaches water level again.

distance covered = speed × time

$= 20 \text{ cm/s} \times \frac{5}{4} \text{ s}$ 2

$= \boxed{25 \text{ cm}}$

■■

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Mathematics Standard
CBSE Examination Paper – 2022
(Term-2)

With Topper's Answer

Time : 2:00 Hours **Max. Marks : 40**

General Instructions:

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **14** questions. **All** questions are compulsory.*

(ii) *This question paper is divided into **three** sections **Sections A, B** and **C.***

(iii) ***Section A** comprises of **6** questions (Q.no. **1** to **6**) of **2** marks each. Internal choice has been provided in two questions.*

(iv) ***Section B** comprises of **4** questions (Q.no. **7** to **10**) of **3** marks each. Internal choice has been provided in one question.*

(v) ***Section C** comprises of **4** questions (Q.no. **11** to **14**) of **4** marks each. Internal choice has been provided in one question. It also contains two case study based questions.*

(vi) *Use of calculator is **not** permitted.*

Set-30/3/3

SECTION – A

Question numbers 1 to 6 carry 2 marks each.

1. In Figure 1, PQ and PR are tangents to the circle centred at O. If $\angle OPR = 45°$, then prove that ORPQ is a square. **2**

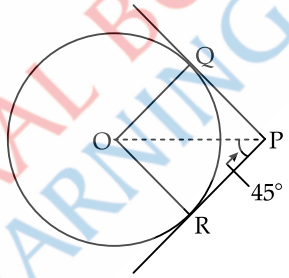


Figure 1

Ans.

To prove :- ORPQ is a square i.e. $\angle O = \angle Q = \angle P = \angle R = 90°$ and $OQ = QP = PR = OR$

Proof :- $\angle OQP = 90°$, $\angle ORP = 90°$ — Tangent is perpendicular to the radius at the point of contact. — (1)

Now, In $\triangle ORP$ and $\triangle OQP$

$OP = OP$ (common)

$OR = OQ$ (Radii of same circle)

$PR = PQ$ (Tangents from an external point to a circle are equal)

Therefore, $\triangle ORP \cong \triangle OQP$ by SSS rule

$\therefore \angle OPR = \angle OPQ$ (By CPCT)

$\Rightarrow \angle OPR = \angle OPQ = 45°$

$\Rightarrow \angle OPR + \angle OPQ = 90°$

$\Rightarrow \angle QPR = 90°$ — (2)

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In quad. OQPR

$\angle OQP + \angle QPR + \angle PRO + \angle ROQ = 360°$ (Angle Sum Property)

$90 + 90 + 90 + \angle ROQ = 360°$

$\angle ROQ = 360 - 270$

$\angle ROQ = 90°$ — (3)

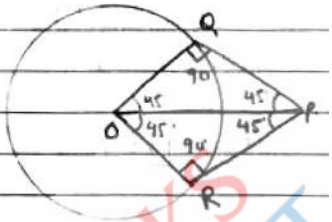
Also $OR = PR$ — (4)

(As $\angle ROP = \angle RPO = 45$) (Isosceles triangle property)

Also $QP = PR$ (Tangents from an external point to a circle are equal) — (5)

$\Rightarrow$ Using (4) and (5)

$OQ = PQ = PR = OR$ — (6)



Using (1), (2), (3), (6)

ORPQ is a square.

Hence proved.

2. Find the mode of the given frequency distribution : **2**

Class	Frequency
15 – 25	6
25 – 35	11
35 – 45	22
45 – 55	23
55 – 65	14
65 – 75	5

Ans.

Class	Frequency
15-25	6
25-35	11
35-45	f_0 22
45-55	f_1 23 → Modal class
55-65	f_2 14
65-75	5

$$\text{Mode} = l + \left\{\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right\} \times h$$

$$= 45 + \left\{\frac{23 - 22}{46 - 22 - 14}\right\} \times 10$$

$$= 45 + \left\{\frac{1}{10} \times 10\right\}$$

$$= 45 + 1 = \boxed{46}$$

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3. Find the common difference '*d*' of an AP whose first term is 10 and the sum of the first 14 terms is 1505. **2**

Ans.

$S_{14} = \frac{n}{2}(2a+(n-1)d)$

$1505 = \frac{14}{2}(2(10)+13d)$

$1505 = 7(20+13d)$

$\frac{1505}{7} = 20+13d$

$215 = 20+13d$

$\frac{195}{13} = d$

$d = 15$

4. For what value of '*n*', are the nth terms of the APs : 9, 7, 5, ... and 15, 12, 9, ... the same ? **2**

Ans.

Quest 9, 7, 5 ——— and 15, 12, 9 ———

$a_n = a+(n-1)d$ $\quad a'_n = a'+(n-1)d'$

$= 9+(n-1)-2$ $\quad = 15+(n-1)-3$

$= 9-2n+2$ $\quad = 15-3n+3$

$= 11-2n$ $\quad = 18-3n$

since $a_n = a'_n$

$11-2n = 18-3n$

$11-18 = -3n+2n$

$-7 = -n$

$\boxed{n=7}$

5. (a) Solve the quadratic equation for x :

$x^2 - 2ax - (4b^2 - a^2) = 0$ **2**

OR

(b) If the quadratic equation

$$(1 + a^2)\,x^2 + 2abx + (b^2 - c^2) = 0$$

has equal and real roots, then prove that :

$$b^2 = c^2\,(1 + a^2)$$ **2**

Ans.(a)

$x^2 - 2ax - (4b^2 - a^2) = 0$

$b^2 - 4ac = 4a^2 - 4[-(4b^2 - a^2)](1)$

$= 4a^2 - 4[-4b^2 + a^2]$

$= 4a^2 + 16b^2 - 4a^2$

$= 16b^2$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{2a \pm \sqrt{16b^2}}{2(1)} = \frac{2a \pm 4b}{2}$

$\Rightarrow x = \frac{2a+4b}{2} \quad x = \frac{2a-4b}{2}$

$x = a+2b \quad x = a-2b$

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6. (a) 150 spherical marbles, each of diameter 1·4 cm, are dropped in a cylindrical vessel of diameter 7 cm containing some water, and are completely immersed in water. Find the rise in the level of water in the cylindrical vessel. **2**

OR

(b) Three cubes of side 6 cm each, are joined as shown in Figure 2. Find the total surface area of the resulting cuboid. **2**

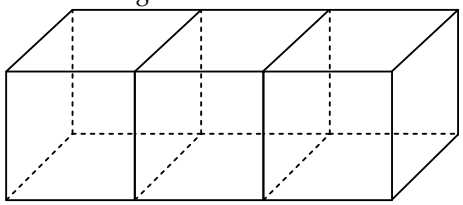


Figure 2

Ans.(b)

Length of cuboid = 18cm
Breadth of cuboid = 6cm
Height of cuboid = 6cm

TSA of the cuboid = $2(lb+bh+hl)$
$= 2(18\times 6+6\times 6+6\times 18)$
$= 2(108+36+108)$
$= 2(252)$
$= 504\ cm^2$

SECTION B

Question numbers 7 to 10 carry 3 marks each.

7. Two boats are sailing in the sea 80 m apart from each other towards a cliff AB. The angles of depression of the boats from the top of the cliff are 30° and 45° respectively, as shown in Figure 3. Find the height of the cliff. **3**

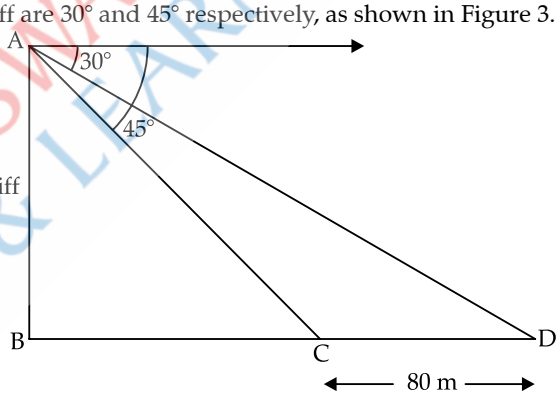
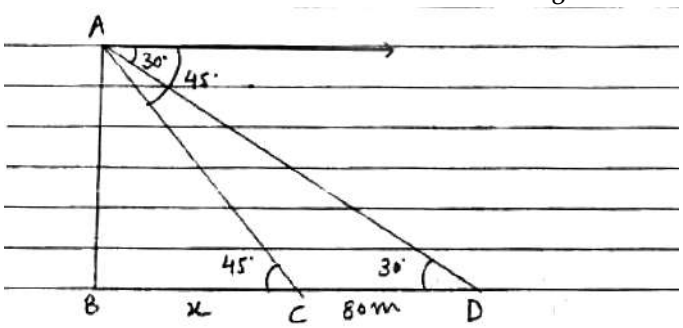


Figure 3

Ans.



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In ΔABC

$\tan 45° = \frac{P}{B} = \frac{AB}{x}$

$1 = \frac{AB}{x}$

$AB = x$

In ΔABD

$\tan 30° = \frac{P}{B} = \frac{AB}{80+x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AB}{80+x}$

$\frac{80+x}{\sqrt{3}} = AB$

$\frac{80+x}{\sqrt{3}} = x$

$80 + x = \sqrt{3}x$

$80 = \sqrt{3}x - x$

$80 = x(\sqrt{3}-1)$

$\frac{80}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = x$

$\frac{80(\sqrt{3}+1)}{(\sqrt{3})^2 - 1^2} = x$

$\frac{80(\sqrt{3}+1)}{3-1} = x$

$\frac{80(\sqrt{3}+1)}{2} = x$

$x = 40(\sqrt{3}+1)$ m

$\Rightarrow AB = 40(\sqrt{3}+1)$ m or 109.2 m

8. For the following frequency distribution, find the median : **3**

Class	Frequency
1400 – 1550	6
1550 – 1700	13
1700 – 1850	25
1850 – 2000	10

Ans.

Class	Frequency	C.F.
1400–1550	6	6
1550–1700	13	19
1700–1850	25	44 → Median class
1850–2000	10	54 = n

$\text{Median} = l + \left(\frac{n}{2} - \frac{CF}{f}\right) \times h$

$= 1700 + \left(\frac{27-19}{25}\right) \times 150$

$= 1700 + \left(\frac{8}{25}\right) \times 150^{6}$

$= 1700 + 48$

$= 1748$

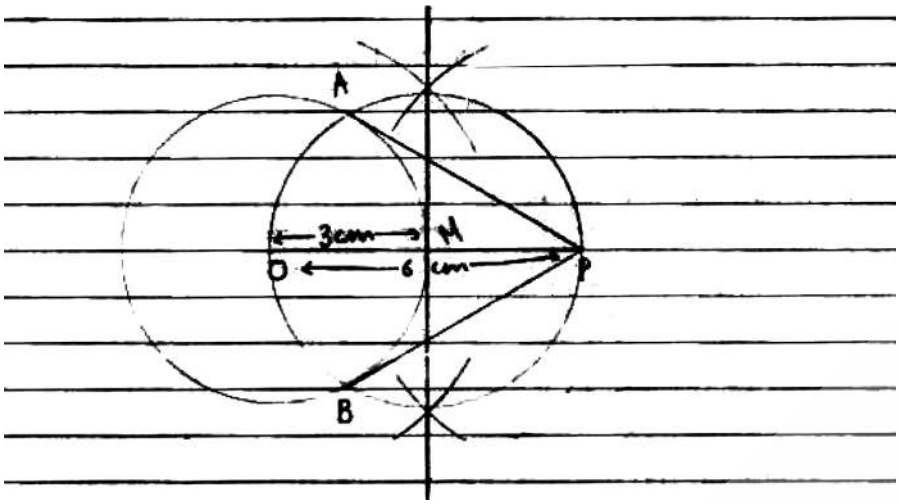
9. (a) Draw a line segment AB of length 8 cm and locate a point P on AB such that AP : PB = 1 : 5. **3**

OR

(b) Draw a circle of radius 3 cm. From a point P lying outside the circle at a distance of 6 cm from its centre, construct two tangents PA and PB to the circle. **3**

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Ans.(b)



Steps of construction →

1. Draw a circle with O as centre and radius 3cm.
2. Take a point P outside the circle at a distance of 6cm from its centre O.
3. Join OP.
4. Construct the perpendicular bisector of OP
5. Name the mid-point of OP as M.
6. Draw a circle with M as centre and radius OM.
7. The second circle intersects the first circle at two points Name them A and B
8. Connect PA and PB.
9. PA and PB are the required tangents.

10. Find the mean of the following frequency distribution : **3**

Class	Frequency
0 – 10	12
10 – 20	18
20 – 30	27
30 – 40	20
40 – 50	17
50 – 60	6

Ans.

Class	Frequency (fi)	xi	fixi
0–10	12	5	60
10–20	18	15	270
20–30	27	25	675
30–40	20	35	700
40–50	17	45	765
50–60	6	55	330
	100		2800

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Mean $= \frac{\Sigma f_i x_i}{\Sigma f_i}$

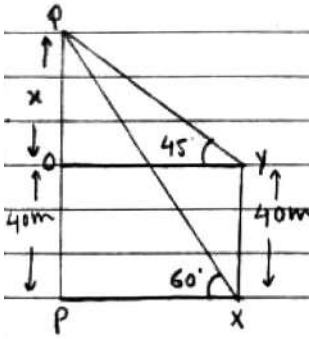
$\bar{x} = \frac{2800}{100} = 28$

SECTION C

Question numbers 11 to 14 carry 4 marks each.

11. The angle of elevation of the top Q of a vertical tower PQ from a point X on the ground is 60°. From a point Y, 40 m vertically above X, the angle of elevation of the top Q of tower PQ is 45°. Find the height of the tower PQ and the distance PX. [Use $\sqrt{3} = 1.73$] **4**

Ans.



In $\triangle QOY$

$\tan 45 = \frac{P}{B} = \frac{QO}{OY} = \frac{x}{OY}$

$1 = \frac{x}{OY}$

$OY = x \Rightarrow PX = x$

In $\triangle QPX$

$\tan 60 = \frac{P}{B} = \frac{QP}{PX} = \frac{40+x}{x}$

$\sqrt{3} = \frac{40+x}{x}$

$\sqrt{3}x = 40 + x$

$\sqrt{3}x - x = 40$

$x(\sqrt{3}-1) = 40$

$x = \frac{40}{(\sqrt{3}-1)} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{40(\sqrt{3}+1)}{(\sqrt{3})^2 - 1^2} = \frac{40(\sqrt{3}+1)}{3-1}$

$x = \frac{40(\sqrt{3}+1)}{2}$

$x = 20(\sqrt{3}+1)$ m

$x = 54.6$ m

$\Rightarrow PX = 54.6$ m

$PQ = 40 + x$

$= 40 + 54.6$

$= 94.6$ m

12. (a) In Figure 4, two circles with centres at O and O of radii $2r$ and r respectively, touch each other internally at A. A chord AB of the bigger circle meets the smaller circle at C. Show that C bisects AB. **4**

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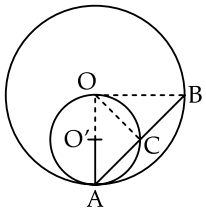


Figure 4

OR

(b) In Figure 5, O is centre of a circle of radius 5 cm. PA and BC are tangents to the circle at A and B respectively. If OP = 13 cm, then find the length of tangents PA and BC. **4**

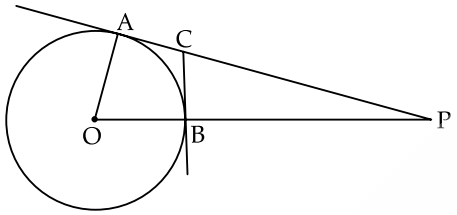
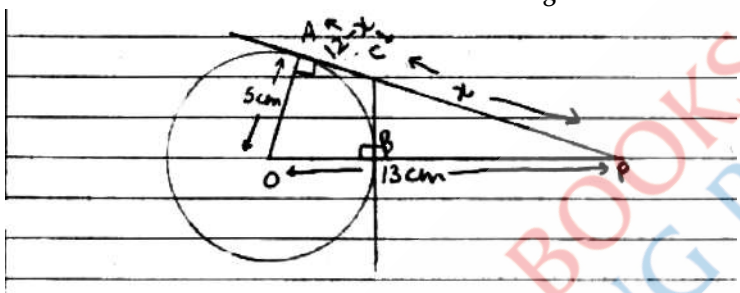


Figure 5

Ans.



$\angle OAP = 90°$ (Tangent is perpendicular to radius at the point of contact)

Using pythagoras theorem in $\triangle OAP$

$H^2 = B^2 + P^2$

$OP^2 = AP^2 + OA^2 \Rightarrow AP^2 = 144$

$13^2 = AP^2 + 5^2$ $\quad AP = \sqrt{144} = 12\text{cm}$

$169 = AP^2 + 25$

$169 - 25 = AP^2$

Let PC be x and AC be $12 - x$

$AC = BC = 12 - x$ (Tangents from point c to the circle are equal)

Also $OP = 13\text{cm}$

$OB = 5\text{cm}$

$BP = OP - OB$

$BP = 8\text{cm}$

$\angle OBC = 90°$ (Tangent is perpendicular to the radius at the POC)

$\Rightarrow \angle CBP = 90°$ (Linear Pair)

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Using pythagoras theorem in Δ CBP

$H^2 = B^2 + P^2$

$CP^2 = BP^2 + BC^2$

$x^2 = 8^2 + (12-x)^2$

$x^2 = 64 + 144 + x^2 - 24x$

$24x = 64 + 144$

$24x = 208$

$x = \frac{208}{24} = \frac{104}{12} = \frac{52}{6} = \frac{26}{3}$

$BC = 12 - x$

$BC = 12 - \frac{26}{3}$

$BC = \frac{36-26}{3} = \frac{10}{3} = 3.34\ m$

Case Study – 1

13. In the picture given below, one can see a rectangular in-ground swimming pool installed by a family in their backyard. There is a concrete sidewalk around the pool of width x m. The outside edges of the sidewalk measure 7 m and 12 m. The area of the pool is 36 sq. m.

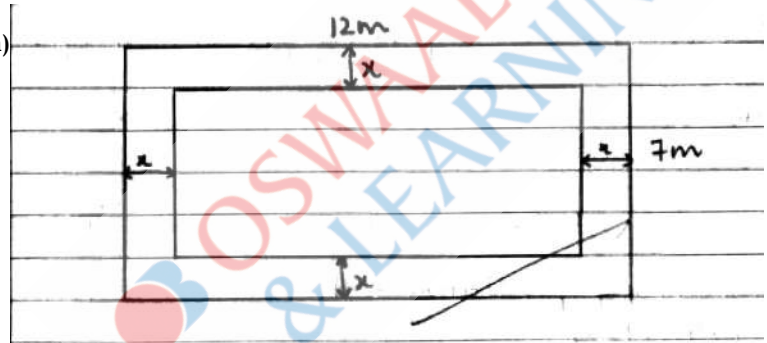


Figure 6

(a) Based on the information given above, form a quadratic equation in terms of x. **2**

(b) Find the width of the sidewalk around the pool. **2**

Ans.(a)



Dimensions of pool — (12-2x) and (7-2x)

Area = length x Breadth

$36 = (12-2x)(7-2x)$

$36 = (84 - 24x - 14x + 4x^2)$

$36 = 84 - 38x + 4x^2$

$36 - 84 = -38x + 4x^2$

$-48 = -38x + 4x^2$

$4x^2 - 38x + 48 = 0 \Rightarrow 2x^2 - 19x + 24 = 0$

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(b) $2x^2-19x+24=0$

$2x^2-16x-3x+24=0$

$2x(x-8)-3(x-8)=0$

$x=\frac{3}{2}$ $x=8$

$x=8$ will be neglected because if $x=8$ the width of sidewall will become more than 16 but they are already given 7m and 12m the dimensions

$\therefore$ $\boxed{x=1.5\text{m}}$

Case Study – 2

14. John planned a birthday party for his younger sister with his friends. They decided to make some birthday caps by themselves and to buy a cake from a bakery shop. For these two items, they decided the following dimensions :

Cake : Cylindrical shape with diameter 24 cm and height 14 cm.

Cap : Conical shape with base circumference 44 cm and height 24 cm.



Based on the above information, answer the following questions :

(a) How many square cm paper would be used to make 4 such caps ? **2**

(b) The bakery shop sells cakes by weight (0.5 kg, 1 kg, 1.5 kg, etc.). To have the required dimensions, how much cake should they order, if 650 cm^3 equals 100 g of cake ? **2**

Ans. (a) CSA of one cap = Aquare cm paper required for one cap

$2\pi x=44$

$2\times\frac{22}{7}\times x=44$

$x=\frac{44\times7}{22\times2}=7\text{cm}$

$l^2=l^2+x^2$

$l^2=24^2+7^2$

$l^2=576+49$

$l^2=625$

$l=25\text{cm}$

CSA of one cap $=\pi rl$

$=\frac{22}{7}\times7\times25$

$=550\ cm^2$

CSA of four caps i.e. square cm paper required for four caps = 4×550
$= 2200\ cm^2$

(b) Volume of cake $= \pi r^2 h$ | $d = 24\ cm$, $r = \frac{d}{2} = 12\ cm$

$= \frac{22}{7} \times 12 \times 12 \times 14.2$

$= 6336\ cm^3$

$650\ cm^3 = 100\ g$ or $0.1\ kg$

$6336\ cm^3 = \frac{0.1\ kg \times 6336}{650} = \frac{6336}{6500} = \frac{95.9}{100} = 0.95\ kg$

So they should order a 1 kg cake.

■■

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